

How we ran IO500 on the cluster

io500-isc25_v2 · standard mode · POSIX · 2026-03-16 · 1 h 46 m wall

Client topology

- Host `sc003` , SLURM partition `htc` , QOS `public`
- **10 nodes × 4 tasks/node = 40 MPI ranks**
- 2 h wall limit, `--export=NONE` (clean env)
- `module load openmpi/4.1.5`
- IOR / mdtest **4.1.0+dev**, `pfind` for find phase

Storage under test

- **POSIX** API for every phase (no MPI-IO, no DAOS)
- Filesystem visible to clients: **3722 TiB, 78 % full**
- No Lustre stripe tuning — `setup()` left at defaults
- Stonewall **300 s**, `stoneWallingWearOut=1` (equal work per rank)

Submission

```
sbatch slurm_script # archives ini + script
# inside the job:
srun ./io500 config.ini \ # 40 ranks
--timestamp $TS
```

Result + datafile dirs are timestamped per run; the script
tar-balls `results/<ts>/` at the end.

Workload — 14 phases, fixed order

1. `ior-easy-write` — file-per-proc, 2 MiB xfer, 9.46 TiB/rank
2. `mdtest-easy-write` — 1 M files/rank, unique dirs
3. `timestamp` ← anchor for `find`
4. `ior-hard-write` — shared file, **47 008 B** xfer
5. `mdtest-hard-write` — 1 M files/rank, 3 901 B each
6. `find` — `pfind` , queue 20 k, work-stealing, hashing
7. `ior-easy-read`
8. `mdtest-easy-stat`
9. `ior-hard-read`
10. `mdtest-hard-stat`
11. `mdtest-easy-delete`
12. `mdtest-hard-read`
13. `mdtest-hard-delete`
14. `ior-rnd4K-easy-read` (informational, not scored)

Artifacts produced

`results/<ts>/` → `result.txt` , `result_summary.txt` , per-phase
`*.txt` + `*.csv` , archived `config.ini` + `slurm_script` , `io500-sc003-<ts>.tgz`

Score formula (for reference): `TOTAL = √(BW × MD)` , geometric mean across the 12 scored phases — reported in `result_summary.txt` . Single successful run on this date; no run-to-run variance captured.